

Flexibility Method Accelerated by H-matrices

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November 2025

Flexibility method accelerated by $\mathcal{H}$ -matrices

Pressure p-0, displacement p-1

Element formulation

To be more aligned with the BEM and potentially with the LBB, we can assign pressure degrees of freedom p_i (or tractions $\mathbf{t}_i$) to element centers and displacement degrees of freedom $\mathbf{u}_j$ to nodes. Then the number of pressure DOFs will be defined by the number of faces and the type of problem $m = d_p \cdot N_e$, where $d_p = 1$ for frictionless problems and $d_p = \dim$ for frictional problems. The number of displacement DOFs is given by the dimension and the number of nodes on the surface of interest $n = \dim \cdot N_n$.

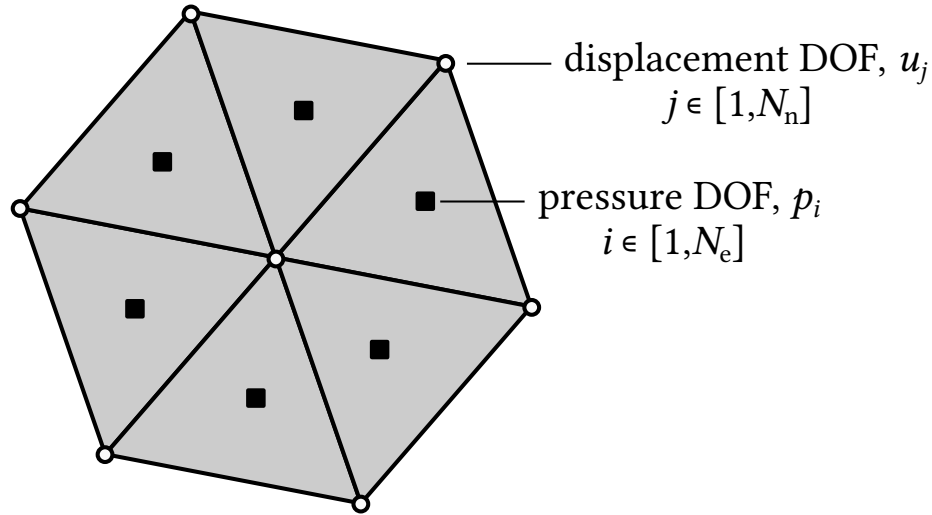


Figure 1: Elements and degrees of freedom

Matrix construction

Within this formulation the flexibility matrix $S_c \in \mathbb{R}^{n \times m}$ is not square. Then the following relation is consistent:

$$[g]_n = [S_c]_{n \times m} [p]_m + [g_0]_n,$$

where $[p]$ is the array of pressure DOFs, and $[g]$, $[g_0]$ are the arrays of gap and initial gaps defined at nodes.

To construct the matrix S_c , we use, for example, the direct sampling method, which consists in applying pressure p_0 at individual elements one-by-one and by measuring induced displacement. It is done by solving FE problem:

$$[K][u]^i = [f]^i,$$

where $[f]^i$ represents the right hand side obtained through integration

$$l(\mathbf{v}) = \int_{\text{el}_i} p_0 \mathbf{n} \cdot \mathbf{v} dS,$$

as shown in the figure below.

The S_c 's matrix i -th row is then given by

$$S_{c_{i,.}} = \frac{1}{p_0} [u]^i.$$

In practice, PETSc solver allows to assemble all required $[f]^i$ in a single matrix $[F]_{N_{\text{DOF}} \times m}$, where N_{DOF} is the total number of DOFs of the FE mesh. Then, by solving the following system in batch mode:

$$[K]_{N_{\text{DOF}} \times N_{\text{DOF}}} [X]_{N_{\text{DOF}} \times m} = \frac{1}{p_0} [F]_{N_{\text{DOF}} \times m},$$

we obtain directly a matrix $[X]$ whose entries at surface nodes can be easily extracted to obtain $[S_c]_{n \times m}$ matrix.

Linear Complementarity Problem

Since the displacement resulting from pressures p is given by:

$$u = S_c p$$

then the gap g change compared to its initial value is given by:

$$g = u + g_0 = S_c p + g_0.$$

The Linear Complementarity Problem (LCP) can be formulated as:

$$g = S_c p + g_0$$

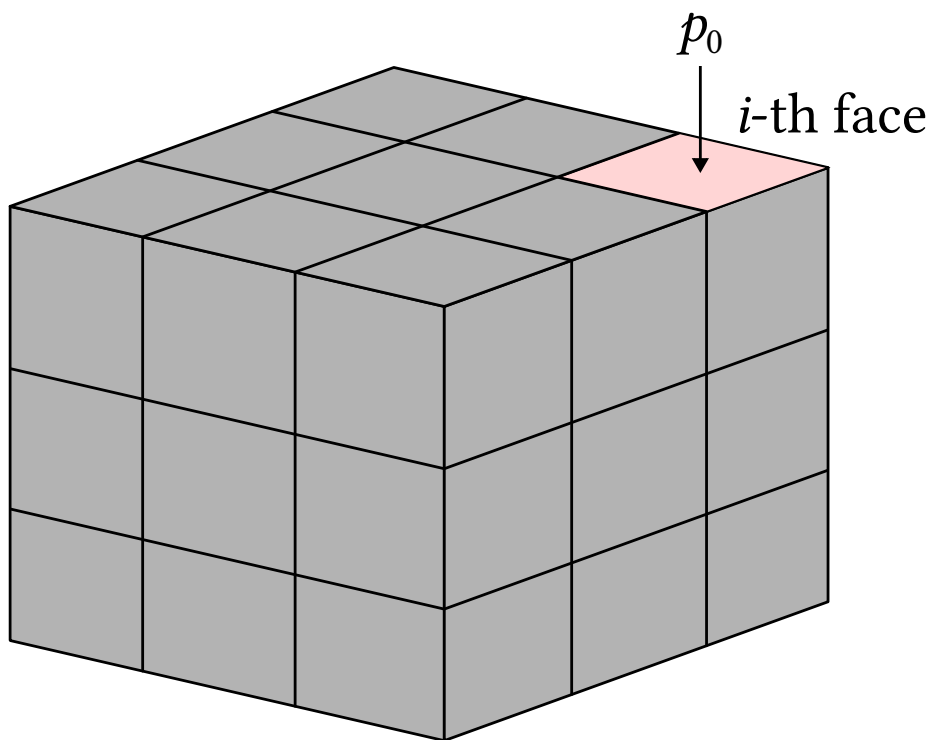


Figure 2: Direct sampling matrix construction

$$g \geq 0, \quad p \geq 0, \quad g \perp p,$$

the last condition, called complementarity condition is not easy to verify when g and p are defined on different spaces, i.e. we cannot simply set $p^\top q = 0$.

The associated quadratic programming problem, for the same space for p and g can be defined as

$$\begin{aligned} \text{minimize } \mathcal{F}(p) &= \frac{1}{2} p^\top S_c p + p^\top g_0 \\ \text{subject to } p &\geq 0 \end{aligned}$$

Again, the last term of the functional cannot be readily evaluated if p and g, g_0 are defined on different spaces.

However, instead of handling the problem

$$g = S_c p + g_0,$$

by multiplying it by $S_c^\top$, we can transform it into

$$S_c^\top M g = S_c^\top M S_c p + S_c^\top M g_0,$$

where M is the mass matrix. This form can be obtained by enforcing the complementarity condition in a weak form:

$$\langle p, g \rangle = \int_{\Gamma_c} p g \, dS = \sum_i p_i \sum_j g_j \int_{\Gamma_c} N_j \, dS = p_i M_{ij} g_j.$$

It could be further generalized by assuming different shape functions for the pressure Φ_i and for the displacement N_j as

$$\langle p, g \rangle = \int_{\Gamma_c} p g \, dS = \sum_i p_i \sum_j g_j \int_{\Gamma_c} \Phi_i N_j \, dS = p_i M_{ij} g_j.$$

Thus weak complementarity $p^\top M g = 0$ uses the correct geometric inner product on Γ_c .

Symmetric formulation

Define

$$H = S_c^\top M S_c, \quad b = S_c^\top M g_0, \quad \lambda = M g.$$

Then the weak equilibrium equation becomes

$$\lambda = H p + b.$$

The complementarity conditions transform to

$$p \geq 0, \quad \lambda \geq 0, \quad p \perp \lambda,$$

which is a proper LCP in the *pressure space*.

The operator H is symmetric positive semidefinite.

Equivalent quadratic program

The LCP (H, b, p, λ) is associated with the convex quadratic program

$$\min_{p \geq 0} \mathcal{F}(p) = \frac{1}{2} p^\top H p + b^\top p.$$

This problem is well posed because:

- $S_c^\top M S_c$ is symmetric,
- the surface mass matrix M supplies a consistent L^2 -pairing between nodal and elemental quantities,

Solution methods

I didn't find good solvers to solve the LCP problem directly, only NNLS (`scipy.optimize.nnls`), but I didn't succeed to make it work properly.

Trying to work with the Quadratic Programming problem directly using different solvers (`quadprog.solve_qp`, `cvxopt.solvers.qp`, `scipy.sparse.osqp`) with forced symmetrization and Tikhonov smoothing

$$H_s = \frac{1}{2}(H^\top + H) + \lambda I$$

but I never could obtain a reliable and smooth result; the solution is very sensitive to λ parameter and the conditionning of the operator H is too bad. Plan to switch to PDAS (primal-dual active-set).